



DESCRIPTION

ALZ 3920 is a small diameter version of our standard submersible hydrostatic level transmitter. High-quality silicon piezoresistive sensor, robust Ø21 mm stainless steel housing and a hydrometric cable with the ventilation tube provide high performance in a slimmer form factor for a wide variety of industrial level measurement applications. The 21 mm probe diameter allows installation in confined spaces such as boreholes. An integrated Pt100 temperature sensor is optionally available.

SPECIFICATIONS

Level: 1 mH₂O up to 250 mH₂O
Basic accuracy: ±0.25% of span
Probe: Ø21 mm, AISI 316L stainless steel
Output: 4...20 mA
Sensor: silicon piezoresistive
Media temperature: -20...+70 °C
Optional: Ex ia

APPLICATIONS

Confined spaces level measurement

- Water, wastewater
- Fuels, oils
- Tanks, wells
- Pump protection

TECHNICAL SPECIFICATIONS

MEASURING RANGES

Pressure range, bar	Level, mH ₂ O	Overpressure, bar	Pressure range, bar	Level, mH ₂ O	Overpressure, bar
0...0.1	0...1	1	0...2.5	0...25	6
0...0.16	0...1.6	1	0...4	0...40	15
0...0.25	0...2.5	1	0...6	0...60	15
0...0.4	0...4	1	0...10	0...100	30
0...0.6	0...6	3	0...16	0...160	60
0...1	0...10	3	0...25	0...250	60
0...1.6	0...16	6	-	-	-

PERFORMANCE

	P > 0.4 bar	P ≤ 0.4 bar
Accuracy, % of span*	≤ ±0.25 (standard) / 0.2 (optional)	≤ ±0.5 (standard)
Temperature effect (% of span / 10 °C)	≤ ±0.1	≤ ±0.2
Compensated range	-20...+70 °C	0...+70 °C
Power supply effect	≤ ±0.05% of span / 10 V	
Load resistance effect	≤ ±0.05% of span / kOhm (transmitters with current output)	
Long-term stability	≤ ±0.1% of span / year	
Response time (10...90%)	≤ 1 ms	

* Accuracy includes non-linearity, hysteresis and non-repeatability.

OPERATING CONDITIONS

Medium temperature (depends on seal)	-20...+70 °C		
Cable service temperature (static applications)	PVC:	-20...+70 °C	
	PUR:	-50...+70 °C	
Storage temperature	-20...+70 °C		
Approval	0Ex ia IIC T6...T4 Ga X		
Temperature class	T4	T5	T6
Ambient temperature	-20...+70 °C	-20...+60 °C	-20...+50 °C
Vibration resistance	10 g RMS, 25–2000 Hz		
Shock resistance	100 g / 11 ms		
Service life	> 100 x 10 ⁶ cycles		

MECHANICAL SPECIFICATIONS

Housing	stainless steel 316L (1.4404)
Seal	FKM; EPDM; NBR
Diaphragm	stainless steel 316L (1.4435)
Cable external jacket	PVC, PUR, FEP
Ingress protection	IP69K

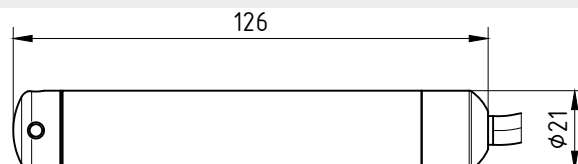
ELECTRICAL SPECIFICATIONS

Output signal	Power supply, U _s	Load resistance, R	Power consumption
4...20 mA / 2-wire	12...36 V	≤ [(U _s - 12 V) / 0.02 A] Ohm*	≤ 26 mA
Safe values for intrinsically safe design 0Ex ia IIC T6...T4 Ga X:			
Maximum voltage, U _i			28 V
Maximum current, I _i			93 mA
Maximum power, P _i			660 mW
Maximum internal inductance, L _i			10 μH
Maximum internal capacitance, C _i			15 nF

ELECTRICAL CONNECTIONS / PIN ASSIGNMENT

Circuits		Cable gland wires
2-wire	power +	white (red)
	power -	brown (blue)
	ground	yellow-green
Pt100 (optional)	T+	yellow
	T-	pink (white)
	T-	gray (green)

DIMENSIONS (mm)



ORDERING CODE

ALZ 3920		-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
UNIT OF MEASUREMENT									
	bar	B							
	kg/cm ²	S							
	mH ₂ O	W							
	kPa	H							
	Other (specify when ordering)	X							
UPPER RANGE LIMIT (URL)									
bar, kg/cm ²	mH ₂ O	kPa							
0.1	0100	1	1000	10	1001				
0.16	0160	1.6	1600	16	1601				
0.25	0250	2.5	2500	25	2501				
0.4	0400	4	4000	40	4001				
0.6	0600	6	6000	60	6001				
1	1000	10	1001	100	1002				
1.6	1600	16	1601	160	1602				
2.5	2500	25	2501	250	2502				
4	4000	40	4001	400	4002				
6	6000	60	6001	600	6002				
10	1001	100	1002	1000	1003				
16	1601	160	1602	Other	XXXX				
25	2501	250	2502						
Other	XXXX	Other	XXXX						
ACCURACY									
	0.25% (P > 0.4 bar) (standard)			C					
	0.5% (P ≤ 0.4 bar) (standard)			D					
	0.2% (P > 0.4 bar)			B					
	Other (specify when ordering)			X					
OUTPUT SIGNAL									
	4...20 mA / 2-wire (standard)			A					
	Other (specify when ordering)			X					
SEALS									
	FKM			F					
	NBR			N					
	EPDM			E					
	Other (specify when ordering)			X					
VERSION									
	Standard			00					
	Pt100 temperature sensor (incompatible with RS-485 / Modbus RTU)			0T					
	Other (specify when ordering)			XX					

ORDERING CODE (CONTINUED)

	ALZ 3920	-X	-XXXX	-X	-X	-X	-XX	// X	-XXXX
CABLE EXTERNAL JACKET									
							PVC	P	
							PUR	U	
							FEP	T	
						Other (specify when ordering)		X	
CABLE LENGTH									
									Any length in meters (e.g., 010M means 10 meters)
									XXXXM
Example: ALZ 3920-W-6000-D-A-F-00 // P-010M									

ACCESSORIES

				
Anchoring clamp for submersible transmitter	PZ 1024 Power supply unit	BZ 05 / BZ 10 Dry air junction box for submersible transmitters		